

COSC363 Assignment 1 Report

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Declaration

I declare that this assignment submission represents my own work (except for allowed material provided in the course), and that ideas or extracts from other sources are properly acknowledged in the report. I have not allowed anyone to copy my work with the intention of passing it off as their own work.

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The theme for the scene is an 'Alien Theme Park' which is set on a flat, disc-shaped planet. The planet is covered in dirt and moss, with rocks and trees scattered around it. In the centre is a spaceship pendulum ride, which consists of a large T-shaped pole (metal textured) with two spaceships connected to each end by cables. The spaceships swing simultaneously in opposite directions, and can hold two aliens each at a time. Two aliens can be seen moving around, getting onto the spaceships which rise up and swing back and forth, before lowering back down to the ground and continuing to move around. There are four other aliens who are stationary in the scene, but continuously wave their arms up and down, their antennas left and right and their legs back and forth. There is also a pink orb hovering around the scene which acts as a spotlight, projecting pink and purple light onto the ride.





Extra Features Implemented:

- **3.1** Planar shadows cast by at least two objects (ride, aliens, rocks, trees)
- **3.2** Complicated pathing (physics based pendulum for the ride)
- **3.3** Aliens get on or off one of the theme park rides without teleporting
- **3.4** A sky box/sphere
- **3.5** A dynamic spotlight: attached to a moving object (pink circle) and changes colour over time (pink to purple)
- **3.6** Smooth user movement: all four controls work smoothly

No diagrams or sketches were used to design the models. They were designed through trial and error in OpenGL where if something was changed, the program was recompiled to view the changes.

Equations Used to Design Surfaces and Animations:

acceleration = $-(g/L) * \sin(\text{angle})$	Used for the rides pendulum swing
angle = amplitude * $\sin(\text{frame})$	Used for the aliens arm, leg and antenna movement
$x = \text{radius} * \cos(\text{angle})$ $z = \text{radius} * \sin(\text{angle})$ $\text{angle} = (2 * \pi * i) / n$ $i = \text{index}, n = \text{number of points}$	Used for the spotlights circular orbit movement and to place 12 circles evenly around the rim of each spaceship
$dx = \text{targetX} - x$ $dz = \text{targetZ} - z$ $\text{distance} = \sqrt{dx^2 + dz^2}$ $x += (dx/\text{distance}) * \text{speed}$	Used to get the aliens to walk towards a target (the ride seats)

Control Functions:

Up Arrow	Moves camera closer
Down Arrow	Moves camera further away
Left Arrow	Rotates camera left
Right Arrow	Rotates camera right
w	Moves camera up
s	Moves camera down
a	Moves camera left
d	Moves camera right
q	Turns wireframe on and off

Build Instructions:

The program was developed on a Windows platform using Visual Studio. To compile and run it on the Linux CSSE lab machines:

1. Navigate to the project directory in a terminal
2. Make sure the textures folder is in the same directory as the source files
3. Compile with the command:
`g++ Main.cpp Models.cpp -o AlienThemePark -lGL -lGLU -lglut -lm`
4. Run the program using:
`./AlienThemePark`

References:

Textures:

- Sky: https://polyhaven.com/a/citrus_orchard_road_puresky
- Ground: https://polyhaven.com/a/coast_sand_rocks_02
- Pole: https://polyhaven.com/a/corrugated_iron_02

Equations:

- Physics pendulum equation for ride: <http://labman.phys.utk.edu>
- Parametric circle equations for spotlight orbit: <https://www.mathopenref.com>
- Sinusoidal wave equation for aliens movement: <https://www.mathsisfun.com/algebra>
- Unit vector and normalisation equations to get the aliens walking towards the target: https://en.wikipedia.org/wiki/Unit_vector

Course Material:

- The 'loadBMP.h' file was reused from Lab 3

Statement of Acknowledgement:

I acknowledge the use of Claude AI for explaining the usage of computer graphics functions and equations to aid movement in my scene. It was also used to answer questions about debugging when I couldn't pinpoint the issue for the program not running.